

Week 02

Basic SQL Operations

Cảm ơn thầy Trần Duy Quang đã cung cấp template cho môn học
Cảm ơn thầy Hồ Tuấn Thanh và thầy Đỗ Nguyên Kha đã xây dựng nội dung thực hành



Lecturer: **TS. Lê Thị Nhàn.**
Lab TA: **ThS. Nguyễn Hải Đăng.**

Notes

Put **ALL** SQL statements to **ONE** text (SQL) document names student ID 1651xxx_yy.sql (yy is a self-grading mark 01-10):

- Put your Student ID and Full name inside and at the beginning of the file as a comment.
- Run ALL lines in the file without any error
- USE the CORRECT database

1

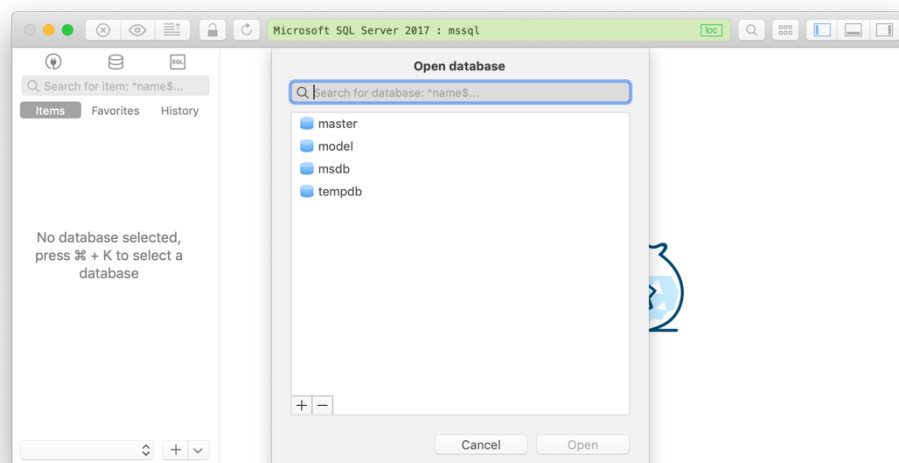
Create a database

The following tutorial is on TablePlus, you can use other tools to do this.

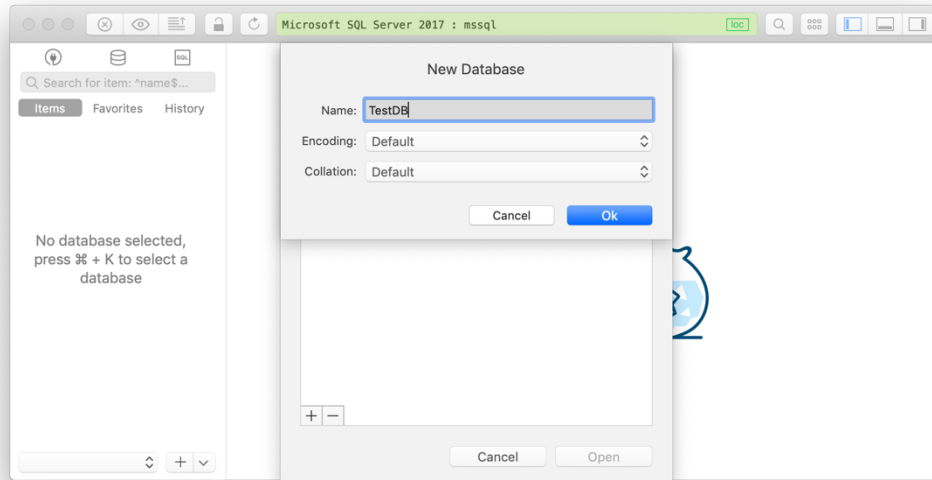
Click the database icon (3 cylinders)



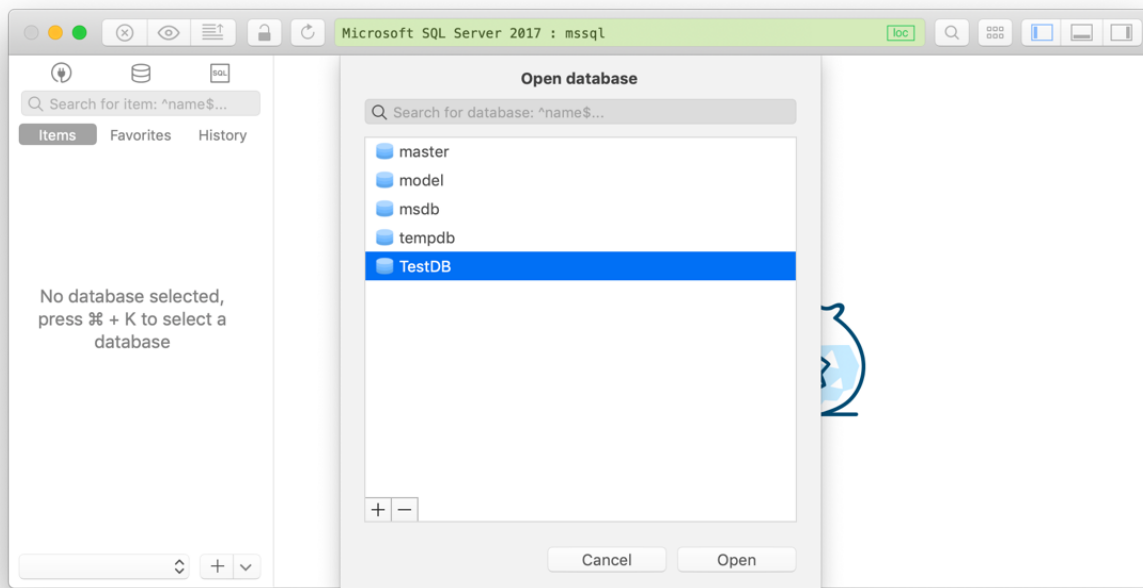
Click the plus icon to create a new database



Enter the new database's name. It should have a simple name without any special character (including spaces).

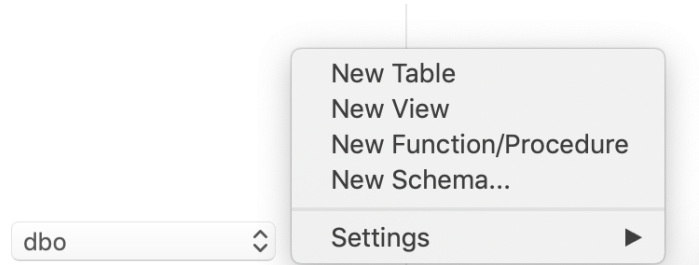


Click **OK** then **Open** the new database in dialog

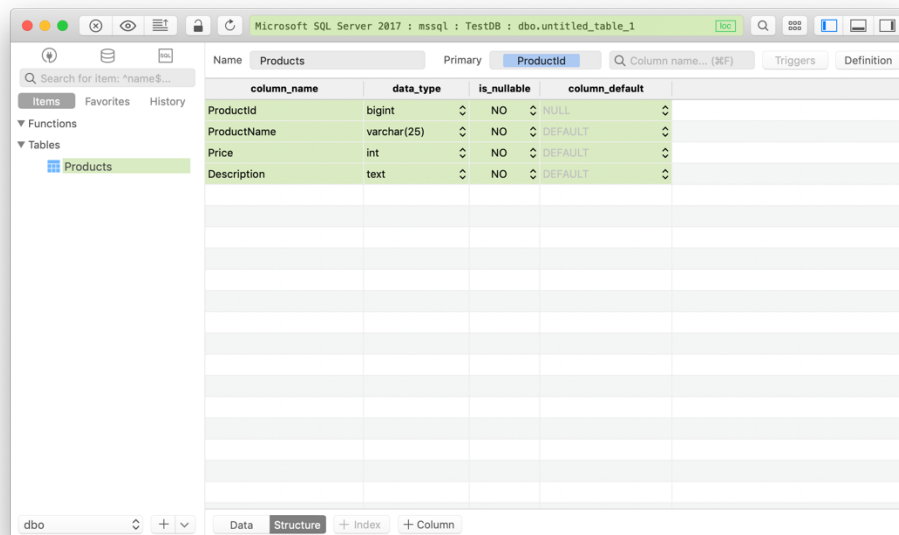


Create a table

Click the down arrow icon at the bottom of the sidebar and select “New Table” or use the plus icon



Enter the table name, column name and data type.



Then click the commit button (Control/Command + S) – the button with the up arrow on the top

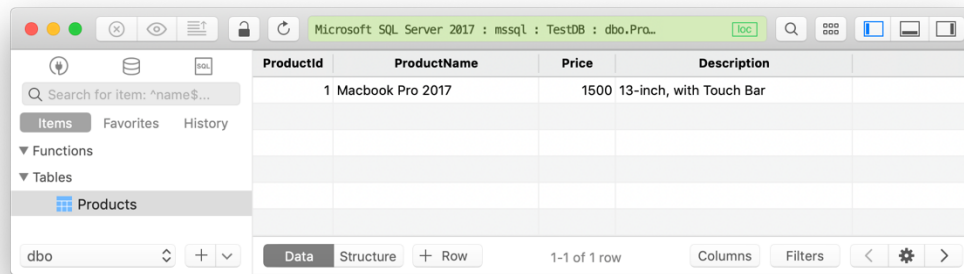
Drop table/database

To drop the table or database (*) just right click on it then choose “Delete” or use the Delete key. Remember to commit the change by clicking the commit button.

(*) see the force remove table by SQL if you cannot drop the database

Insert data

Change the bottom tab to **Data** to insert data to the table row by row then use the commit button to save the changes.



Microsoft SQL Server 2017 : mssql : TestDB : dbo.Pro...

Search for item: ^name\$...

Items Favorites History

▼ Functions

▼ Tables

Products

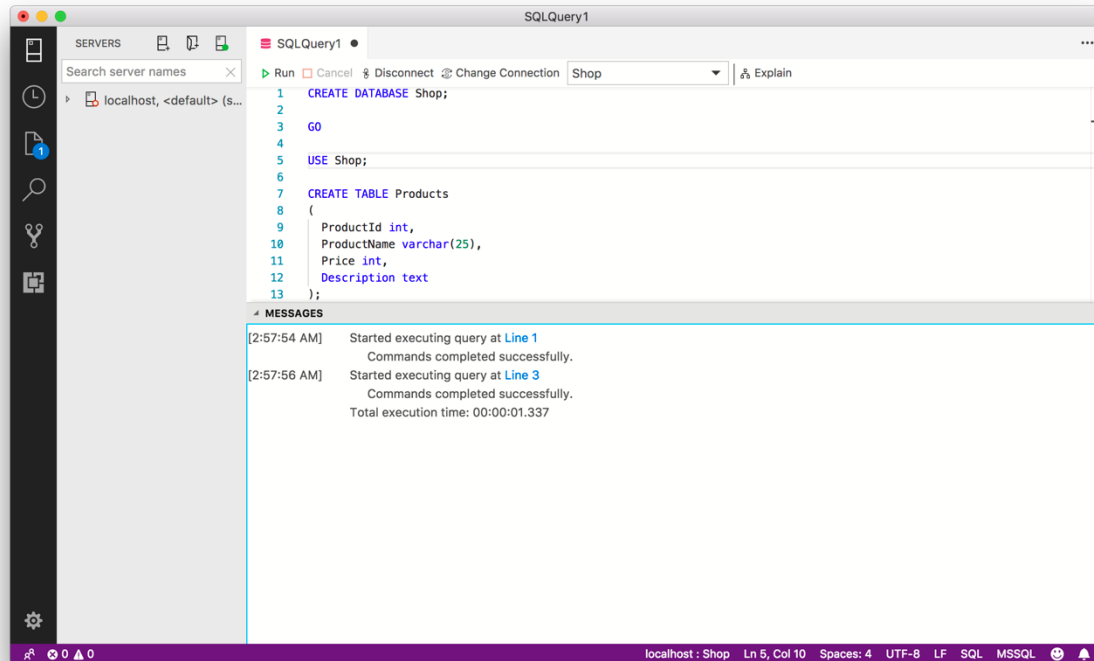
dbo

ProductId	ProductName	Price	Description
1	Macbook Pro 2017	1500	13-inch, with Touch Bar

Data Structure + Row 1-1 of 1 row Columns Filters

2

Use SQL editor



3

Database operations

Create database

```
CREATE DATABASE [database name];
```

Example:

```
CREATE DATABASE TestDB;
```

SQL keywords are case insensitive

```
create database TestDB;
```

Remove database

```
DROP DATABASE [database name];
```

Example:

```
DROP DATABASE TestDB;
```

Select database

```
USE [database name];
```

Example:

```
USE TestDB;
```

Force remove database

```
-- Rollback all transactions
ALTER DATABASE Shop SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
-- Select master database
USE master;
-- Remove database
DROP DATABASE Shop;
```

Table operations

Create table

```
CREATE TABLE [table name]
(
    [attribute 1] [data type 1],
    [attribute 2] [data type 2],
    ...
);
```

Example:

```
CREATE DATABASE Shop;
GO
USE Shop;

CREATE TABLE Products
(
    ProductId int,
    ProductName varchar(25),
    Price int,
    Description text
);
```

Remove table

```
DROP TABLE [table name];
```

Example:

```
DROP TABLE Products;
```

Data operations

Insert

```
INSERT INTO [table name] (column 1, column 2,...) VALUES (value 1, value 2);
```

Example:

```
INSERT INTO Products (ProductId, ProductName, Price, Description) VALUES (1, 'Macbook Pro 2017', 1500, '13-inch, with Touch Bar');
```

To insert Vietnamese with accent marks or diacritics (Unicode) use the **N** prefix and **nvarchar** data type

```
INSERT INTO Products (ProductId, ProductName, Price, Description) VALUES (1, 'Váy  
tụt Chi Pu', 25, N'Đóa hoa hồng (QUEEN)');
```

Query

```
SELECT column 1, column 2... FROM [table name];
```

Example:

```
SELECT ProductId, ProductName, Price, Description FROM Products;
```

or use *

```
SELECT * FROM Products;
```

Delete

```
DELETE FROM [table name] WHERE [conditions];
```

Example:

```
DELETE FROM Products WHERE ProductId = 1;
```

Assignments

Database name: University

Tables:

- Students:
 - StudentId: varchar(7)
 - FirstName: varchar(25)
 - LastName: varchar(25)
 - Age: int
 - Class: varchar(10)
- Courses:
 - CourseCode: varchar(10)
 - CourseName: varchar(255)
 - Year: int
 - Semester: int
 - Lecturer: varchar(64)

Create SQL file with requirements:

1. Create database
2. Create tables
3. Add sample data, at least **40** students and **5** courses. Use students information in your class to create sample data.
4. Query all students
5. Remove 1 student and 1 course
6. Remove all tables
7. Remove database

1. Lab

Requirement 1, 2 and partial requirement 3 (**10** students and **2** courses)

Submission: StudentID_W02_NumCompleteProblems.sql

YY: Number of complete problems. YY: 03

2. Homework

All requirements

Submission: StudentID_W02_YY.sql

YY: Number of complete problems. YY: 07